

MERN STACK INTERNSHIP – DAY 16 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

Name: Chayan Soni

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FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

Internship Day 16 – Product Review & Rating Module Development

Objective

The objective of Day 16 was to design and implement a Product Review & Rating Module for the MERN E-Commerce application. The task involved creating MongoDB schemas for product reviews and ratings, developing REST APIs for review management, calculating average product ratings dynamically, integrating review functionality into the React frontend, validating user inputs, and testing all features before pushing the updated code to GitHub.

Tasks Performed

1. Created Product Review and Rating Schemas

Designed MongoDB schemas using Mongoose to manage user reviews and product ratings efficiently.

Product Review Schema Fields

- User ID
- Product ID
- User Name
- Review Comment
- Rating Value

- Created At
- Updated At

Rating Information

- Rating values restricted between 1 and 5 stars.
- Each review linked to a specific user and product.
- Maintained data consistency through schema validation rules.

The schema structure ensured efficient storage and retrieval of review-related information.

2. Developed Review Management REST APIs

Implemented RESTful APIs to handle complete review operations within the application.

Add Review API

Purpose: Allow authenticated users to submit reviews and ratings for products.

Features:

- Accepts review comment and rating.
- Validates user input.
- Stores review data in MongoDB.
- Associates review with the selected product.

Edit Review API

Purpose: Enable users to update previously submitted reviews.

Features:

- Verifies review ownership.
- Updates review comments and ratings.
- Saves modified information to the database.
- Returns updated review details.

Delete Review API

Purpose: Allow users to remove their reviews.

Features:

- Authenticates user requests.
- Removes review from the database.
- Updates product rating calculations accordingly.

Fetch Product Reviews API

Purpose: Retrieve all reviews associated with a product.

Features:

- Returns complete review list.
- Displays reviewer information.
- Shows ratings and review comments.
- Supports frontend review rendering.

3. Implemented Average Rating Calculation

Developed backend logic to calculate product ratings dynamically.

Features

- Computes average rating based on all submitted reviews.
- Updates automatically whenever a review is added, edited, or deleted.
- Displays accurate rating information for products.
- Improves product credibility and user experience.

Example Calculation

Average Rating = Total Rating Sum ÷ Number of Reviews

This ensured that product ratings always reflected current user feedback.

4. Frontend Integration

Integrated review and rating functionality into the React frontend application.

Implemented Features

- Review submission form.
- Star rating selection component.
- Review editing functionality.

- Review deletion option.
- Product review display section.
- Average rating display on product pages.

The frontend communicates with backend APIs using Axios to provide a seamless user experience.

5. Input Validation and Error Handling

Implemented validation mechanisms on both frontend and backend.

Client-Side Validation

- Required review comments.
- Rating value validation.
- Form field validation before submission.

Server-Side Validation

- Authentication checks.
- Review ownership verification.
- Invalid rating prevention.
- Database error handling.

Error Handling Features

- User-friendly error messages.
- API response validation.
- Graceful handling of unexpected failures.

This improved application security, reliability, and user experience.

6. API Testing Using Postman

All APIs were thoroughly tested using Postman.

Tested APIs

1. Add Review API
2. Edit Review API
3. Delete Review API
4. Fetch Product Reviews API

Validation Performed

- Successful API execution.
- Authentication verification.
- Input validation testing.
- Error response testing.
- Database record verification.

All APIs responded correctly and performed as expected.

Technologies Used

- MongoDB
- Express.js
- React.js
- Node.js
- Mongoose
- Axios
- Postman
- Git & GitHub

Challenges Faced

- Designing relationships between products and reviews.
- Maintaining accurate average rating calculations.
- Implementing secure review modification controls.
- Synchronizing frontend review updates with backend data.
- Handling validation and error scenarios effectively.

Learning Outcomes

Through this task, I learned:

- Designing review management database schemas.
- Building scalable REST APIs for review systems.
- Implementing rating aggregation logic.
- Managing user-generated content securely.
- Integrating backend APIs with React frontend components.

- Performing comprehensive API testing using Postman.
- Implementing client-side and server-side validation techniques.

API Documentation Summary

Add Review

Method: POST

Purpose: Add a new product review and rating.

Edit Review

Method: PUT

Purpose: Update an existing review.

Delete Review

Method: DELETE

Purpose: Remove an existing review.

Fetch Product Reviews

Method: GET

Purpose: Retrieve all reviews for a specific product.

GitHub Submission

The Product Review & Rating Module was successfully implemented, tested, documented, and pushed to the GitHub repository.

Screenshots Section

okay okay

Hello • 19/06/2026

this product wass okay

3.0

Edit

Delete

good

Chayan Soni • 19/06/2026

hello this is a good product

5.0



Furniture

Ergonomic Office Chair

Breathable mesh chair with lumbar support, tilt lock, and adjustable armrests.

Rs. 8,999

Free delivery above Rs. 1000

4.0

2 review(s)

[Viewing reviews](#)

Out of stock

RATINGS & REVIEWS

Ergonomic Office Chair

Customers can add, edit, delete, and view reviews for this product.

No reviews yet. Be the first to review this product.

0.0

average
rating

0

total reviews

Furniture

category

Review title

Summarize your experience

Rating

5 stars

Review comment

Describe product quality, comfort, fit, or
delivery experience

Add review

MongoDB Compass - localhost:27017/day15_order_management/reviews

Connections Edit View Collection Help

Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- config
- day15_order_management
 - orders
 - products
 - reviews
 - users
- day3db
- hirespec
- jwt-auth
- local
- mern-day6
- product-management
- wishlist-management

reviews

localhost:27017 > day15_order_management > reviews

Documents 1 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

[Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#)

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```
{
  "_id": ObjectId('6a346eb5514cf39fd1b80a70'),
  "product": ObjectId('6a346071c9cfc3641bb691b0'),
  "user": ObjectId('6a345d7209287dc84f032fe1'),
  "userName": "Chayan Soni",
  "title": "good",
  "comment": "hello this is a good product",
  "rating": 5,
  "createdAt": 2026-06-18T22:18:29.882+00:00,
  "updatedAt": 2026-06-18T22:18:29.882+00:00,
  "__v": 0
}
```

All the screenshots are from locally running the project .